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Course Name : Natural language processing

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CIA-1

1. The regular expression is :

a. \b(0[0-9][2][1-9]/[1([0-9][1-9]/3[19]{3}\b

+91 02030 → mobile number invalid

+91 630032 → mobile number valid

03-11-2026 → Date is valid

3-11-26 → Date is invalid

gmail.1923.cm → email is invalid

@gmail.com → email is valid

c.

Line	Appointment	Mobile number	Date	valid / invalid
L_1	05/03/2025 (match)	NONE	NONE	valid
L_2	9876542303	valid	NONE	valid
L_3	ravi.k@college	None	none	invalid
L_4	None	NONE	5/3/25	invalid

(6.) Indian mobile number

\b[?:-;0-6][6-9]?!+91]){3}

+91 6300359268 → valid

+91 0652329893 → invalid

The mobile number or Date is formate of LI is YYYY-MM-dd

2026-11-03 is valid.

* The regular expression is helps to extract the structure formate of in unstructured.

(2.)

Given

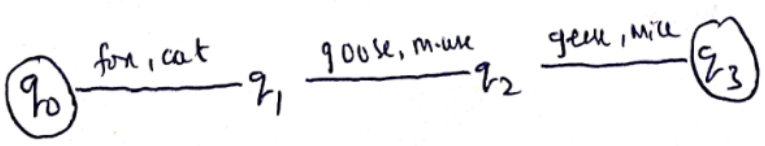
The lexican contains

reg-noun = { fox, cat, dog }

irreg-sg-noun = { goose, mouse }

irreg-pl-noun = { geese, mice }

(a.) Finite state automaton have the $q_0, q_1, q_2, q_3 \cdots q_n$



q_0 → Starting state

q_3 → finite state (end state)

(b.)

S.NO	morpheme	Accept / Reject	Reason
1	dog + -s	Accept	follow irregular noun
2	mouse + -s	Reject	irregular noun
3	geese	Accept	regular plural noun
4	fox + -s	Accept	irregular noun
5	mice + -s	Reject	irregular singular noun

(c.) lexical finite-state morphologies

regular Noun = fox, cat, dog

irregular singular noun = goose, mouse

irregular plural noun = geese, mice

(d.) Differentiate inflention and derivational Morphology.

Walked, unhappiness, computerization, geese

Walked = Walk + ed

unhappiness = un + happy + ness

Computerization = computer + ize + ation

geese = gee + se

3. (i) poster stemmer

Given words

ponies, motoring, relational, hopefulness and Generalization

ponies

ponies → ies

↓

ponie

↓

poni

motoring

motoring → ing

↓

motorin

↓

motori

↓

motor

relational

relational → ional

↓

relat

hopefulness

hopefullness → fullness

↓

hope

Generalizations

generalizations → izations

↓

general

(ii) sentences :

<s> the cat sat </s>, <s> the cat ran </s> and <s>

the dog sat <s>.

Unigram

corpus	count
The	3
cat	2
sat	1
ran	4
dog	2

Bigram

corpus	count
The	4
cat	6
sat	2
ran	8
dog	4

$$P(w_i/w_{i-1}) = \frac{P(w_{i-1}/w_i)}{C(w_{i-1})}$$

$P(w_i/w_{i-1}) = 1/2$ $(\because S_1 = \text{"the cat sat"})$

$P(w_i/w_{i-1}) = 1/4$ $(\because S_2 = \text{"the dog sat"})$

$P(w_i/w_{i-1}) = 1/4$ $(\because S_3 = \text{"the dog ran"})$

(q) Given Sentence

<s> I like tea </s>, <s> I like coffe </s> and

<s> I drink tea </s>

$\therefore V = 6$

$N = 12$

(a) Unigram

corpus	counts
like	4
tea	3
coffe	2
drink	2

Bigram

Corpus	count
Tea	2
coffe	8
like	4
drink	3

$p(\text{tea}|\text{like}) = 2/4 = 1/2$

$p(\text{drink}|\text{like}) = 3/4$

(b.) compute add-on (laplace)

C* of "like tea"

(c.) compute katz-style

$d = 0.5$

α = like

$V = 6$

$N = 12$ (<s>)

left over-mass back-off weight = like = $2/4 = 1/2$

(d.) given $\alpha_2 = 0.7$

$\alpha_2 = V/N = 6/12 = 1/2$

(5.) Treebank Target

PRP, MD, VB, NN

PRP → personal pronoun

MD → middle verb

VB → form of verb

NN → Noun

Given

(a) HMM (Hidden marcobmorphology)

P(PRP) = 0.5, P(NN) = 0.3, P(VB) = 0.1, P(MD) = 0.1

Transitions → PRP → MD = 0.4, PRP → MB = 0.5

Emmisions: P(They | PRP) = 0.8 ; P(can | MD) = 0.3

P(can | NN) = 0.01 , P(fish | NN) = 0.02 , P(fish | VB) = 0.01

taggar (can → MD) = 0.65

taggar(fish → NN) = 0.02 / 0.01

= 0.1